

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. in CSE Final Examination, 2022
Course Code: CSE-2103 Course Name: Digital Electronics & Pulse Technique
Total Marks: 70 Time: 3 Hours

Answer any 5 (Five) of the following Questions

1. a) Define leakage current and Knee voltage. Implement universal gate by diode. 2+3
b) Show that Transistor acts as a switch. 4
c) Implement Exclusive NOR gate by transistor. 5

2. a) Describe Bipolar logic family; Explain NOR gate using TTL 3+3
b) Define Digital IC; Describe the Implementation of NAND gate using RTL 2+3
c) Describe the implementation of NAND gate using DTL 3

3. a) What is counter modulus? Design a 4 bit synchronous counter circuit with counting sequence. 5
b) Differentiate between latches and flip flop. Describe Clocked SR flip-flop. 2+3
c) Implement Right shift register circuit with working procedure. 4

4. a) Prove that, Transistor works as a switch 4
b) Draw the internal circuit diagram of 555 timer circuit and describe the functionality of each pin. 5
c) Mention the difference between Clipping and Clamping circuit with example. 5

5. a) What is PLA? Implement the circuit with PLA having the following functions: 5
 $F1(A, B, C) = \sum (3, 5, 6, 7)$
 $F2(A, B, C) = \sum (0, 2, 4, 7)$
b) Describe the pulse transformer circuit and its equivalent circuit. 5
c) Explain pulse transmission with input and output waveforms in different media. 4

6. a) Draw the block diagram of a D/A converter and explain its operation 5
b) Describe negative clipping circuit. 4
c) Write down the usages of a clipping circuit 3
d) Define LED & LCD 2

7. a) Explain the memory Read and Write operation. 4
b) Explain pulse generator block diagram. 5
c) Implement mono-stable multivibrator circuit. 5